

5.2.59

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Question : Solve the system of equations

$$2x + 3y + 3z = 5 \quad (1)$$

$$x - 2y + z = -4 \quad (2)$$

$$3x - y - 2z = 3 \quad (3)$$

Solution :

The system of equations in matrix form is :

$$\begin{pmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 3 \end{pmatrix} \quad (4)$$

Forming the augmented matrix,

$$\left(\begin{array}{ccc|c} 2 & 3 & 3 & 5 \\ 1 & -2 & 1 & -4 \\ 3 & -1 & -2 & 3 \end{array} \right) \quad (5)$$

Using Gaussian elimination,

$$\left(\begin{array}{ccc|c} 2 & 3 & 3 & 5 \\ 1 & -2 & 1 & -4 \\ 3 & -1 & -2 & 3 \end{array} \right) \xrightarrow[R_2 \rightarrow 2R_2 - R_1]{R_3 \rightarrow 2R_3 - 3R_1} \left(\begin{array}{ccc|c} 2 & 3 & 3 & 5 \\ 0 & -7 & -1 & -13 \\ 0 & -11 & -13 & -9 \end{array} \right) \xrightarrow{R_3 \rightarrow R_3 - \frac{11}{7}R_2} \left(\begin{array}{ccc|c} 2 & 3 & 3 & 5 \\ 0 & -7 & -1 & -13 \\ 0 & 0 & \frac{-80}{7} & \frac{80}{7} \end{array} \right) \quad (6)$$

Using back substitution we get :

$$\frac{-80}{7}z = \frac{80}{7} \quad (7)$$

$$z = -1 \quad (8)$$

$$-7y - z = -13 \quad (9)$$

$$-7y + 1 = -13 \quad (10)$$

$$7y = 14 \quad (11)$$

$$y = 2 \quad (12)$$

$$2x + 3y + 3z = 5 \quad (13)$$

$$2x + 3 = 5 \quad (14)$$

$$x = 1 \quad (15)$$

Therefore the solution for the system of equations is :

$$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \quad (16)$$

Intersection of Three Planes and Solution Point P

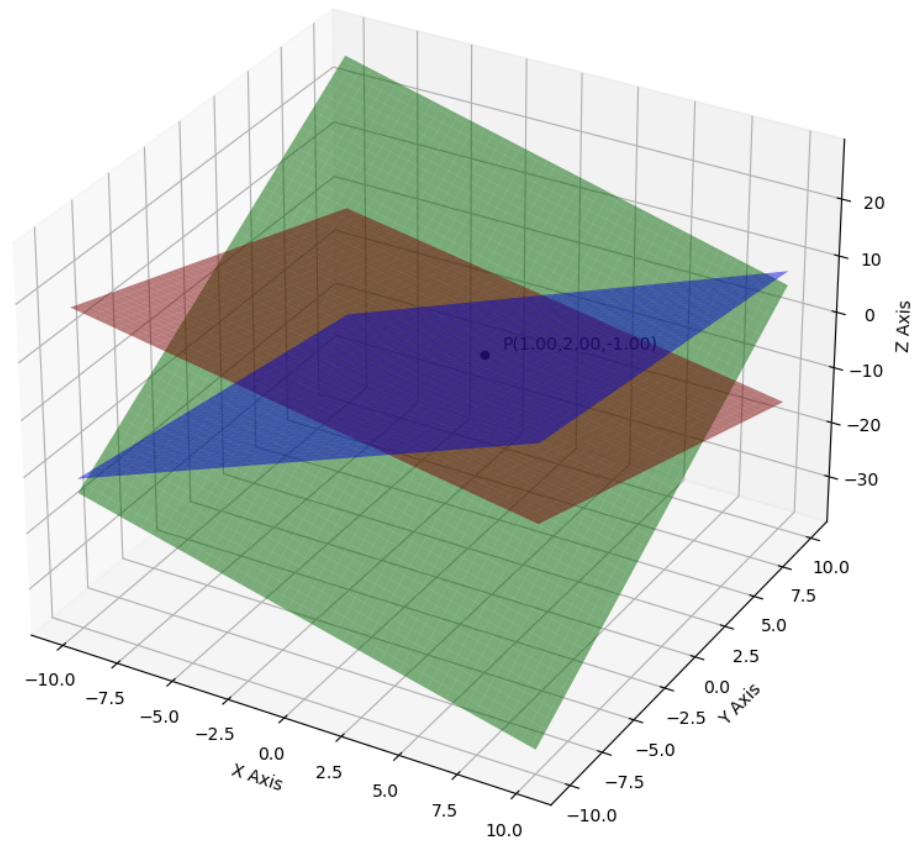


Fig : Planes